Methods for fitting the model

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Abstract

1 Introduction

We assumed that the parameters that were previously used in these models was reasonable to quantitatively explain the cell cycle model (Goldbeter), we have thus a good confidence in their value. However, parameters from Rb E2F switch have been recently studied (Kwon et al.) and thus model verification has been less complete. We also understand that the use of models is strongly dependant on the cell context and thus using several models on cancer cell line may benefit from fitting to our experimental data.

20 parameters from the mitotic module were chosen, they correspond to the 20 parameters the most sensitive to E2F switching (Kwon et al.). Even though these are local sensitivities and rely on the original parameters, we We decided to focus on a global sensitivity analysis of those parameters regarding to the IMT distributions and the fractional proliferation model parameters calculated from the experimental data. Finally we used genetic algorithms to obtain better fits (?)

2 Sobol sequence

The Sobol sequence generation of parameters allowed us to explore the sensitivity of the model regarding 30 his parameters. The sobol sequence has the advantage to fill the space more efficiently compared to picking parameter vectors from uniform distribution [ref]. The advantage is that we can always increase the finesse of the search afterwards in a simple way by generating the sequence of the sequence unlike a simple grid search.

The Sobol sequence were computed in the log scale of the biochemical parameters. The dimensions were centered using the original parameters values, taken from [ref] and range from -2 to +2 the original location in the log space. 120.000 sets of parameters were generated using the Sobol sequence.

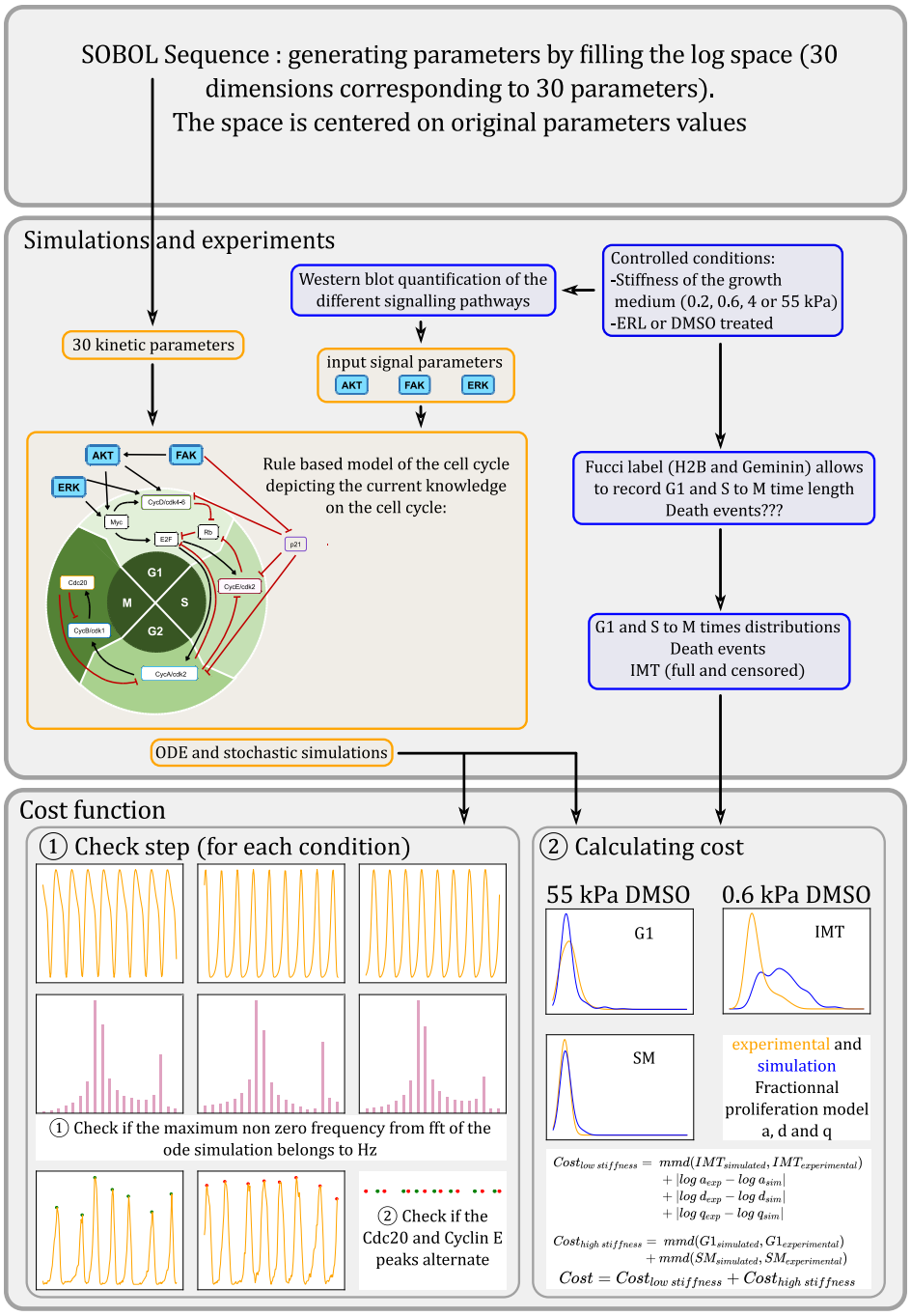


Figure 1: Strategy for parameters sensitivity analysis

3 Cost function design

The cost function aim is to compare both simulations and experiments given a metrics. The cost function is summarized (fig 1).

3.1 Simulation

PySB Stochastic and ode signalling level from WB

3.2 Check step

One issue we had to deal with is that some parameters would not give oscillations and some would make some protein concentration skyrocket which is not possible inside a cell. The check step was thus allowing us to calculate the whole cost for the parameters that are not worth it.
The first part is to see if the concentrations didn't exceed a threshold of 15 $mol.L^{-1}$ with ODE simulation of the model. The second part is to check for oscillations. It was made by checking if the maximum amplitude of the fast fourier transform of the ODE simulation belonged to and Hz^{-1}. Then it was checked that peaks of Cdc20 and Cyclin E alternate in the stochastic simulation using the peakfinder algorithm (ref).

3.3 Cost calculation

The cost was calculated by comparing experiments and data. We used to types of treatment: High stiffness and low stiffness, both DMSO treated. We compared G1 and S to M distributions for the high stiffness condition and we compared IMT distributions and fractional proliferation model parameters for the low stiffness condition.

3.3.1 Comparing parameters of the fractional proliferation modle (ref)

The model takes into account both death and cell cycle as well as the experimentally taken data where a cell that doesn't divide may be into a long cell cycle or in quiescence. It's based on different cell states: quiescence and dividing cells (ref). We translated the code to calculate the parameter from R to Python.

Likelihood of Quiescence model
We find the best parameters for the quiescence model by minimizing the log of the likelihood of fractional cells and quiescent cells given by

$$log(\prod f(x_i, y_j|\mu, \sigma, \lambda)) = log(\prod_i (1-Q) f(x_i|\mu, \sigma, \lambda) \times \prod_j (Q + (1-Q)F(y_j|\mu, \sigma, \lambda))) \tag{1}$$

With x_i are the observed lifespans, these correspond to cycling cells and f gives the probability, given the Exponentially-Modified Gaussian (EMG) model that this intermitotic time happens. And y_j are the censored lifespans, meaning we only observe their birth time during the experiment, thus they may be quiescent cells (with a probability of q) or cycling cells (with a probability of $(1-q)F(y_j|\mu, \sigma, \lambda)$).

Thus we find the best μ, σ, λ and q.

The methods used for minimizing is SLSQP (Sequential quadratic programming) from minimize in the scipy package.

Death rate (s^{-1})
The death rate is calculated with the following formula:

$$a = \frac{Number\ of\ deaths}{Total\ observed\ time}\ (s^{-1}) \tag{2}$$

(In the original paper this is estimated by fitting the log2(proliferation curve))

Division Rate (s^{-1})
The division rate is calculated with the following formula:

$$2\int_0^\infty e^{-d\ \tau} EMG(\tau|\mu, \sigma, \lambda) d\tau = 1 \tag{3}$$

Finding this using brentq algorithm from Scipy (Brent's method to find a zero of the function f) and d as a variable

Quiescence Rate (s^{-1})

$$q = \frac{Q}{1-Q} d \ (s^{-1}) \tag{4}$$

3.3.2 Comparing two distributions using maximum mean discrepancy

The Maximum Mean discrepancy is a metrics used to measure the difference between two distributions $\mathbb{P}$ and $\mathbb{Q}$. It takes advantage of the kernel trick. The MMD is defined as follow : (ref)

$$M_k(\mathbb{P}, \mathbb{Q}) = \|\mu_{\mathbb{P}} - \mu_{\mathbb{P}}\|^2_{\mathcal{H}} \tag{5}$$

$$= \mathbb{E}_{\mathbb{P}}[k(x, x')] - 2\mathbb{E}_{\mathbb{P},\mathbb{Q}}[k(x, y)] + \mathbb{E}_{\mathbb{Q}}[k(y, y')] \tag{6}$$

The mmd function has been taken from a Matlab file (https://gitlab.centralesupelec.fr/2014jinw/GMDA/tree/43585c0f2b0b0e205ac54217a9f2c99521029d53/code)

However, in our case we have a one dimensional data set (set of intermitotic time) and we use the Radial basis function for the kernel. [ref]

4 Results of the fit (results or methods)

2776 sets of parameters out of 120.000 generated by the sobol sequece went through the check step. (fig 4) We plotted the distribution (fig 3à) of the first 100 best parameters. They usually have normal distribution, however, here some of them look flat, which can be explained Finally we plotted the best parameters on the metrics we used to assess the cost. We can see that most of them fit only one condition, only a subset fit roughly both.

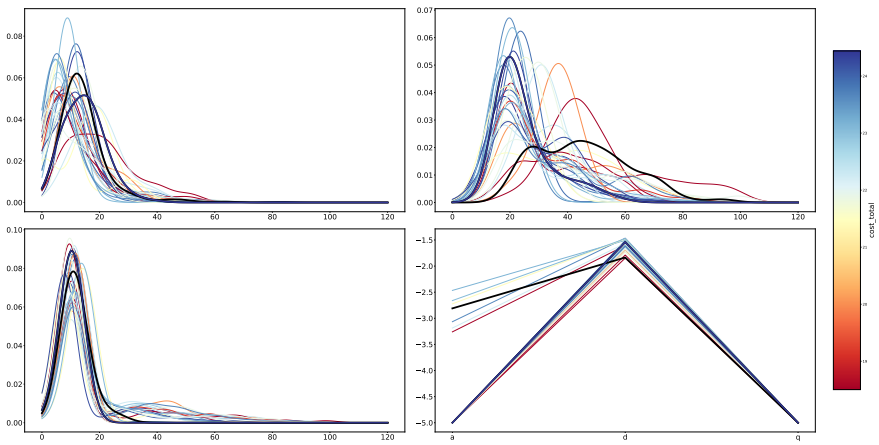


Figure 2: Comparison of fits for the 30 best costs. Black thick line: experimental data, blue thick line: original parameters.

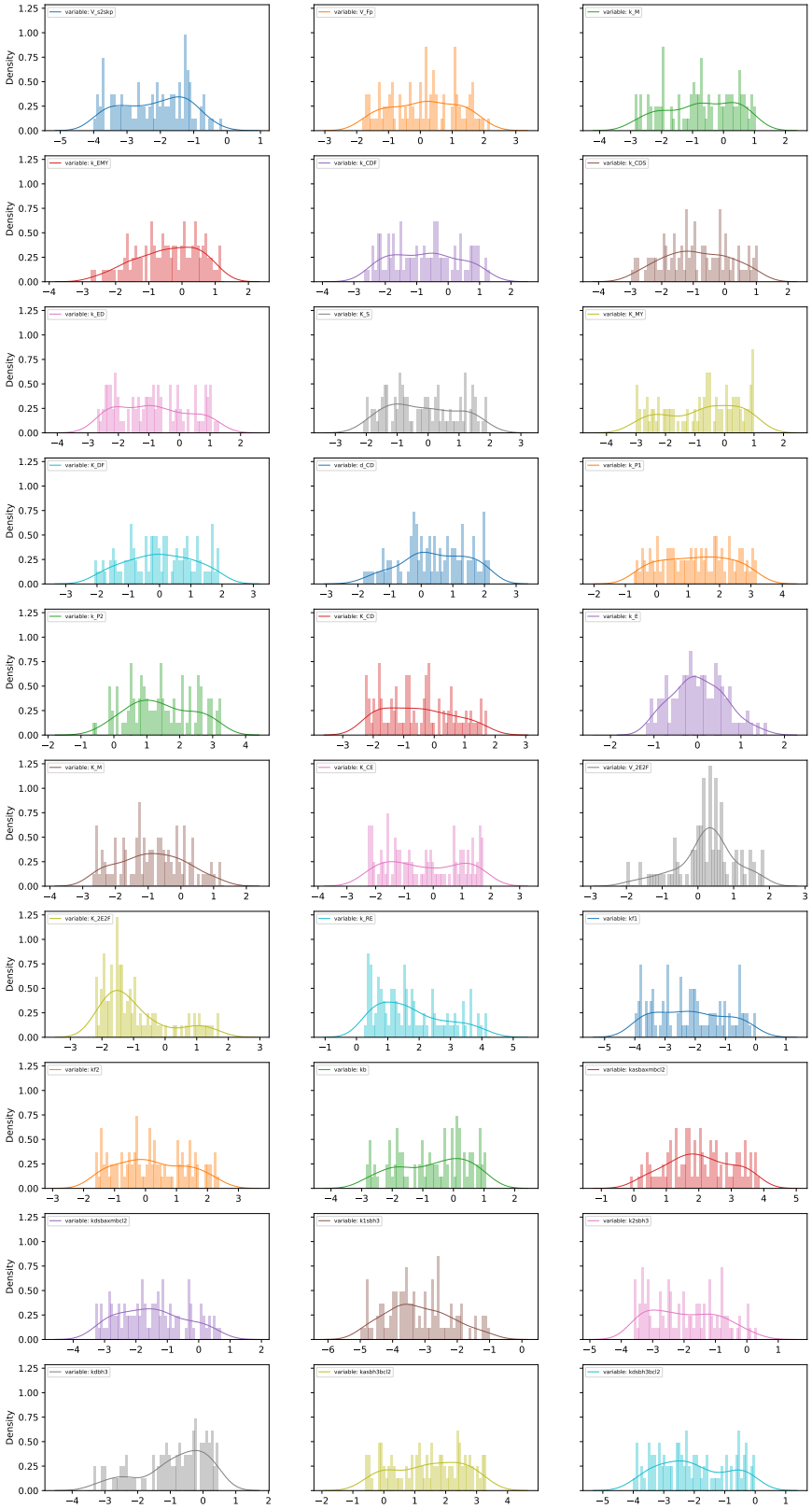


Figure 3: Distribution for 30 parameters for the 100 best sets of parameters

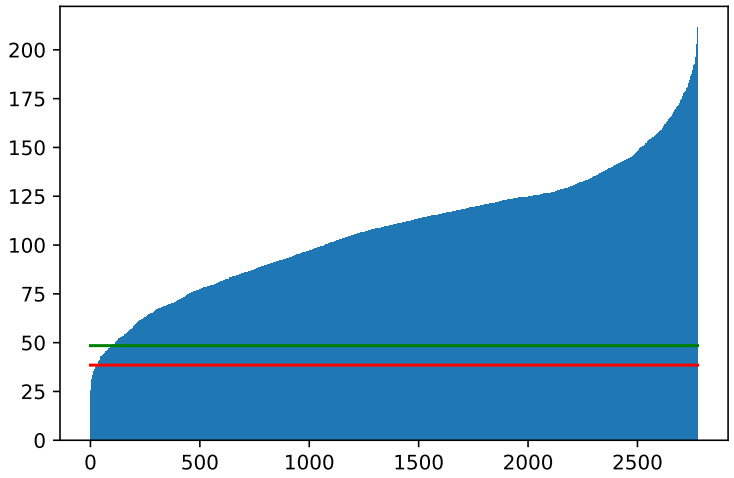


Figure 4: Distribution of the cost. Red line = Original parameters, green line = 100 best costs

5 PSO

The global sensitivity approach allowed us to have an idea about parameter repartition over the space. Hwever they don't give very good fit, especially for the low stiffness condition. Starts from the best points from sobol seq analysis

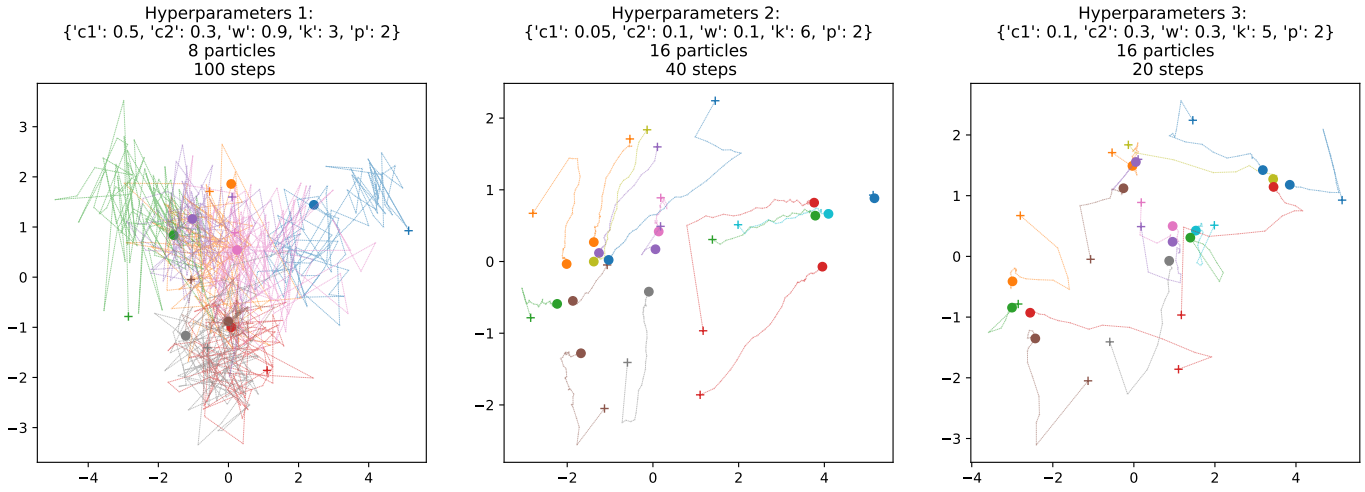


Figure 5: Particle swarm optimization with three different sets of hyperparameters

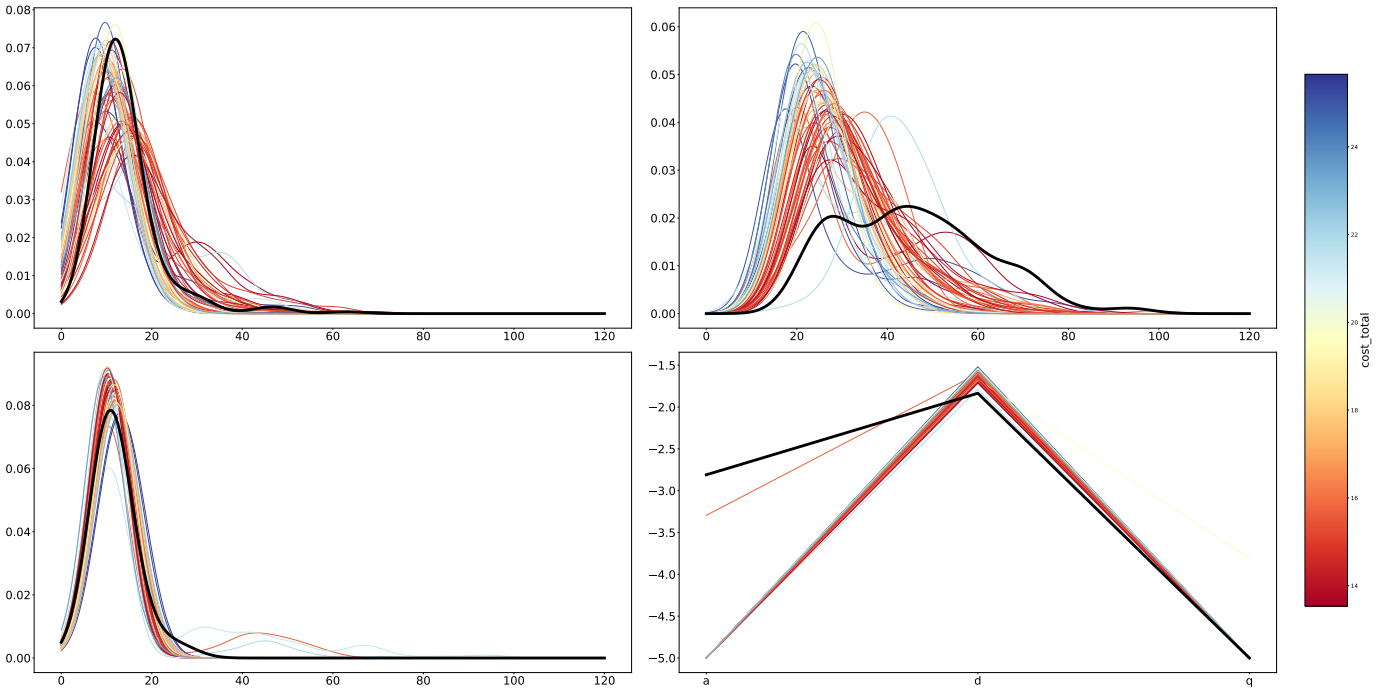


Figure 6: Results for Particle swarm optimization with hyperparameters 1

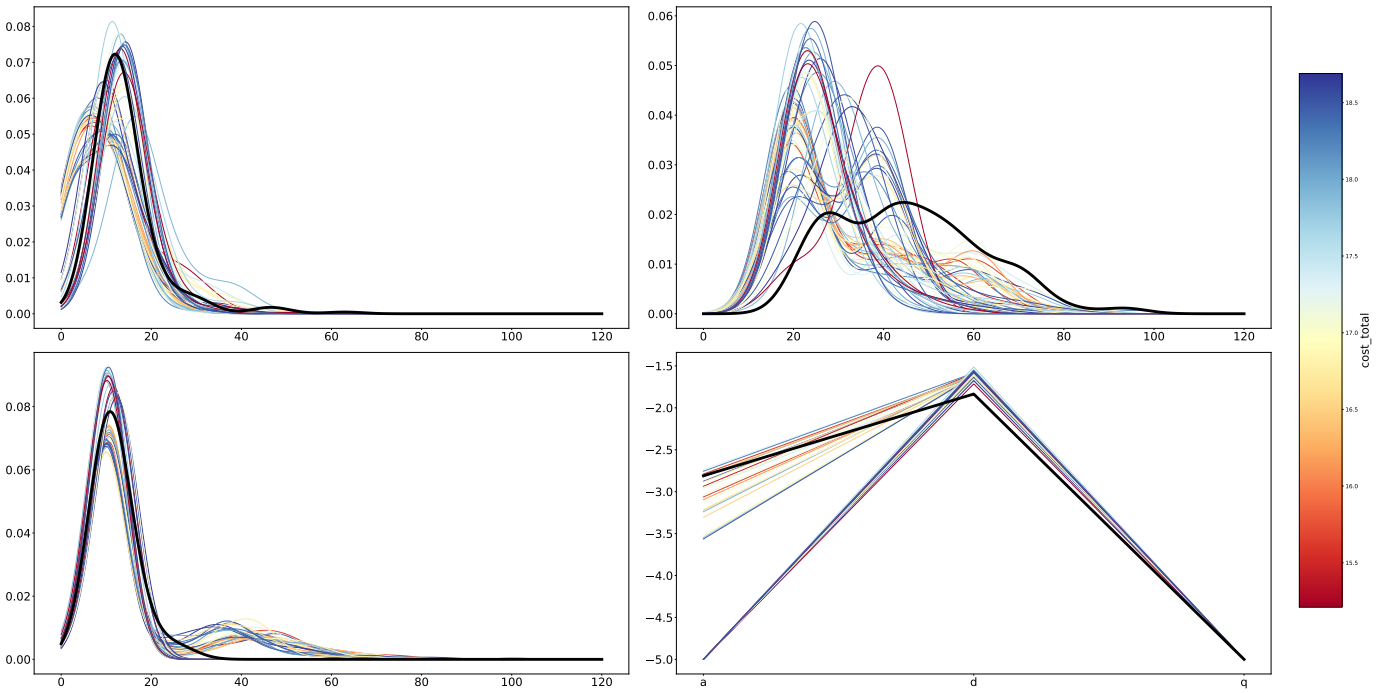


Figure 7: Results for Particle swarm optimization with hyperparameters 2

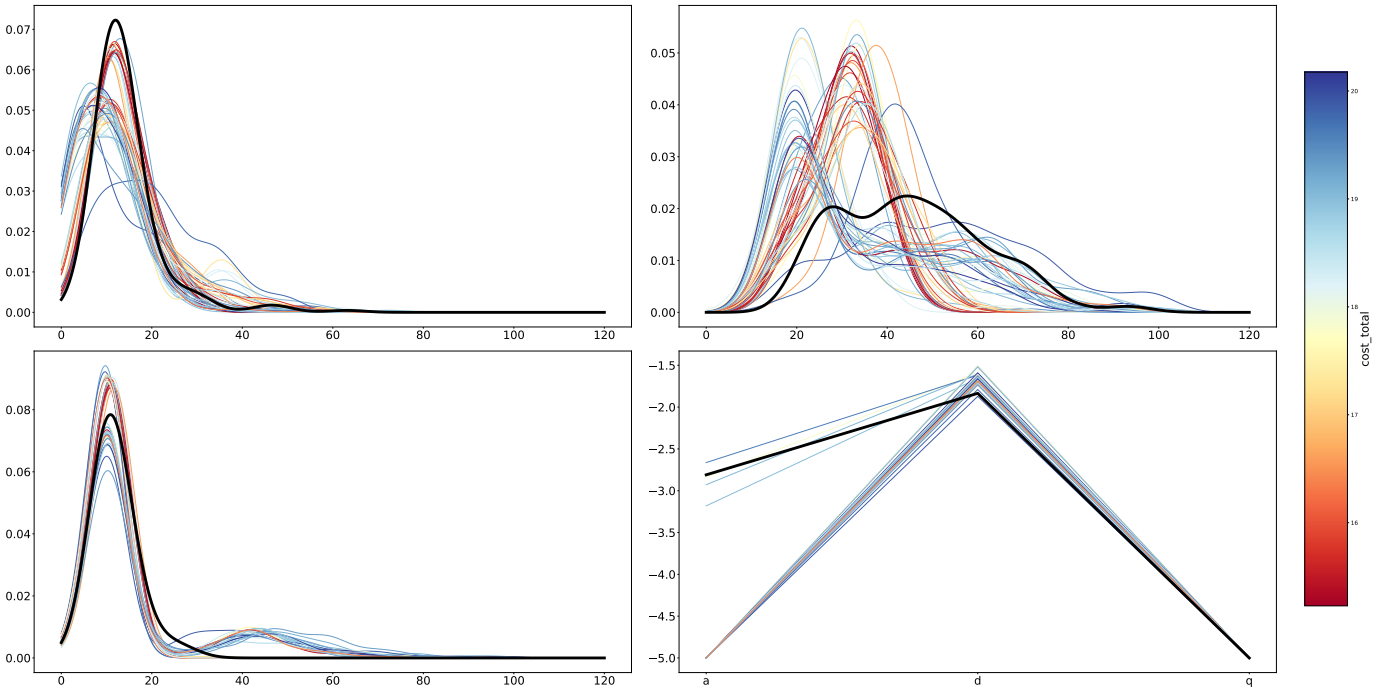


Figure 8: Results for Particle swarm optimization with hyperparameters 3